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Quiz 3

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1. Write the first 3 term of $a_n = 3n+2$, i.e. a_1 , a_2 , and a_3 ? $a_1 = 5$ $a_2 = 8$ $a_3 = 11$

2. Write the 3rd, 4th, and 5th terms of

$$a_1 = 5, a_2 = 7, a_n = n \cdot a_{n-1} + a_{n-2}$$

$$a_3 = 26 \quad a_4 = 111 \quad a_5 = 555 + 26 = 581$$

3. Write an explicit formula for 1, 3, 5, 7, 9,

$$a_n = 2n - 1$$

4. Write a recursive formula for 1, 3, 5, 7, 9,

$$a_1 = 1, \quad a_n = a_{n-1} + 2$$

5. Write a recursive formula for 1, 1, 3, 5, 11, 21, 43, 85,

$$a_1 = 1, \quad a_2 = 1, \quad a_n = a_{n-1} + 2(a_{n-2})$$

6. Let $A = \{1, a, c\}$ $B = \{1, b\}$ $U = \{1, 2, 3, a, b, c\}$

What is $f_A(a)$? 1

What is $f_B(a)$? 0

What is $f_{A \cup B}(a)$? 1

What is $f_{A \cap B}(a)$? 0

7. a) Give the regular set for $(a \vee b)$ $\{a, b\}$
b) Give the regular set for $(ab)^*$ $\{ab, abab, ababab, \dots\}$

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Is $pqrqq$ in the regular set corresponding to the regular expression $(p \vee q) r q^*$? no

Is $abccabc$ in the regular set corresponding to the regular expression $(abc \vee c)^*$? yes

Give the regular expression for the following

9. The set of strings over $\{a, b\}$ in which a occurs exactly once.

$$b^* a b^*$$

10. The set of length two or more over $\{a, b\}$ in which all the a 's precede the b 's.

$$a^* b b^* \vee a a b^* \vee a a^* b b^*$$